



The effects of tax clienteles on disclosure: Evidence from the municipal bond market

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ABSTRACT

The municipal bond market has long been criticized for its lack of disclosure, which prior research often attributes to weak regulatory oversight. We offer another explanation: tax clienteles. Most municipal bonds are tax-exempt, and thus primarily attract high-tax retail investors, who may lack the incentive or ability to demand or monitor disclosures. In contrast, taxable bonds attract a broader investor base, including institutional investors who can demand disclosure. We find that issuers provide more continuing disclosures in years with taxable bonds outstanding relative to years without. Issuers also increase disclosures after issuing their first taxable bond and decrease disclosures after calling their last taxable bond. Exploiting a tax law change, we find that disclosures increase following plausibly exogenous increases in taxable bond issuances. Our results suggest that the tax clienteles for municipal bonds affect issuers' disclosure practices and can inform those concerned with disclosure noncompliance in the municipal bond market.

1. Introduction

In this study, we examine whether tax clienteles affect disclosure in the municipal bond market. The municipal bond market is the largest capital market for state and local governments, with over 50,000 issuers and more than \$4 trillion of municipal bonds outstanding (SIFMA, 2023). Municipal continuing disclosures provide information about the financial and operating conditions of state and local governments, which municipal bond investors can use to assess risk and make asset allocation decisions (Baber and Gore, 2008; Cuny et al., 2025).¹ Yet, the disclosure environment in the municipal bond market has been widely criticized for its lack of transparency. Despite SEC Rule 15c2-12 requiring issuers to provide post-issuance disclosures, many municipalities fail to comply (e.g., Gillette et al., 2020).² Evidence suggests that regulatory efforts to improve continuing disclosures have been ineffective (Maffett et al., 2025). To date, weak regulatory oversight is the primary documented explanation for disclosure noncompliance in the municipal bond market (e.g., Maffett et al., 2025; Rhodes, 2015). We propose and examine another explanation, namely a lack of

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¹ We focus on municipalities' post-issuance continuing disclosures (i.e., financial filings and event notices). We use the term "continuing disclosures" to differentiate from official statements filed along with each new bond issuance.

² Gillette et al. (2020) suggest that nearly 40 percent of issuers do not comply with continuing disclosure requirements. SEC Rule 15c2-12 does provide exceptions to the requirement for post-issuance continuing disclosures, for example, when an issuer's total outstanding debt is less than \$10 million or when an issuer's bonds are sold exclusively to sophisticated investors.

investor demand and monitoring from the primary tax clientele (i.e., high-tax retail investors) for tax-exempt municipal bonds, which constitute the majority of outstanding municipal bonds. Accordingly, we examine whether the taxability of municipal bonds, which can shift a bond's clientele from high-tax retail investors to institutional investors, is associated with issuers' continuing disclosure practices.

Because interest income from most municipal bonds is exempt from federal income tax, high-tax retail investors have a large presence in the municipal bond market. As of 2022, retail investors *directly* held 40.4 percent of outstanding municipal bonds (Bagley et al., 2022).³ Consistent with retail investors forming a primary clientele for tax-exempt municipal bonds, prior research finds that bond prices are sensitive to individual tax rates (e.g., Ang et al., 2010; Green, 1993; Schaefer, 1982). Retail investors tend to hold municipal bonds to maturity and rely on credit ratings to determine the credit quality of issuers (SEC, 2012). In addition, Cornaggia et al. (2022) find that retail investors are less attentive to bond market news relative to institutional investors. As a result, the primary tax clientele for tax-exempt bonds may have limited incentives or ability to demand and monitor disclosures.

Although most municipal bonds are tax-exempt, some are taxable. Municipalities may issue taxable bonds to gain flexibility in the use of proceeds, attract additional capital and a broader investor pool, or respond to tax law changes. Because taxable bonds offer a higher pretax return than tax-exempt bonds of similar risk, they attract a broader set of investors, especially institutional investors who may not benefit from the tax savings from tax-exempt bonds, such as tax-exempt entities (e.g., pension funds or endowments) or entities structured as flow-throughs for tax purposes (e.g., mutual funds or ETFs). Institutional ownership of municipal bonds steadily increased after 2009 and saw a steep increase in 2020 and 2021 (Bagley et al., 2022), which coincides with two waves of taxable municipal bond issuances in response to tax law changes.⁴

Prior research generally supports the view that continuing disclosures are informative for municipal bond market participants.⁵ As the issuance of taxable bonds shifts an issuer's clientele toward institutional investors, the issuer may change its disclosure practices in response to investor demand. Institutional investors price their positions frequently and have a stronger demand for economic and fundamental information than individual investors (e.g., Cuny, 2018).⁶ Institutional investors have been shown to serve a key monitoring role and affect the disclosure practices of public firms (e.g., Abramova et al., 2020; Tsang et al., 2019). If they serve a similar monitoring role in the municipal bond market and represent a primary clientele for taxable municipal bonds, all else equal, we expect issuers to provide more continuing disclosures when they have taxable bonds outstanding. Yet, some may argue that disclosures are not particularly important in the municipal market because secondary market trading is far less active compared to equity markets and municipal disclosures are not very timely. Additionally, while individual direct ownership is still the dominant form of municipal bond investment, individuals are gradually shifting toward indirect ownership via funds and ETFs, thereby increasing institutional ownership for tax-exempt bonds.⁷ These unique institutional features of the municipal bond market, along with historically low default risk, suggest that tax clienteles may not affect issuers' disclosure practices.

To examine our research question, we obtain data on U.S. municipal bonds for fiscal years between July 2009 and December 2022 from the Mergent Municipal Securities Database. We collect issuers' continuing disclosures from the Municipal Securities Rulemaking Board (MSRB). Our sample includes 59,256 unique issuers and 560,652 issuer-years. We focus on issuers' financial filings because these disclosures provide direct information about issuers' financial strength and risk (e.g., Baber et al., 2013; Baber and Gore, 2008). For completeness, we also examine issuers' material event notice filings as well as total continuing disclosures.

We measure the tax clientele for an issuer's outstanding bonds using an indicator variable that captures the existence of at least one taxable bond outstanding at the end of the fiscal year.⁸ The decision to issue a taxable bond is likely endogenous, and the reasons for issuing taxable bonds could also explain why issuers provide more disclosures. We address this concern in three ways. First, we include an extensive set of controls and issuer fixed effects throughout our analyses to help alleviate the concern that issuer characteristics

³ For comparison, retail investors account for less than 20 percent of trading volume in public equities (Hajric and Graffeo, 2021).

⁴ The tax law changes include the establishment of Build America Bonds (BABs) in 2009, and the Tax Cuts and Jobs Act of 2017 (TCJA). We discuss both law changes in greater detail in Section 2.

⁵ For example, studies have found associations between municipal disclosures and credit rating changes (Cuny et al., 2025), borrowing costs (Baber et al., 2013; Reck and Wilson, 2014), secondary market trading and pricing (Cuny et al., 2025; Reck and Wilson, 2006; Sherrill and Yerkes, 2018), and transaction costs (Cuny, 2018).

⁶ For example, a comment letter to the Government Accounting Standards Board (GASB) from State Farm notes, "As municipal bond investors, we rely on the financial statements and notes to the financial statements to provide fair, accurate information on the financial condition of governmental entities as a critical part of the investment decision making process" (https://www.gasb.org/page/ShowPdf?path=RskDisc_ED_CL016.pdf, last accessed March 2025).

⁷ Using macro-level data, Bagley et al. (2022) document a gradual decline in individual investors' *direct* ownership of U.S. municipal debt, from 54 percent in 2004 to 40.1 percent in 2022. Yet ownership by mutual funds, closed-end funds, and ETFs has increased from 16 percent in 2004 to 26 percent in 2022. This increasing trend in institutional ownership, which can affect both tax-exempt and taxable bond issuers, may reduce our ability to detect a tax clientele effect on disclosure.

⁸ We use the taxability of outstanding bonds to capture tax clientele given limited data on municipal bond ownership. While data on individual direct ownership of municipal bonds are available at the macro-level (Board of Governors of the Federal Reserve System, 2025), prior studies (e.g., Cornaggia et al., 2022) use trade size as a proxy for institutional versus individual ownership at the bond-level. This approach could be problematic to the extent that institutional investors break orders into multiple small transactions (Cready et al., 2014). Thus, our approach of using bond taxability to infer clientele is meaningful and can potentially be used in future research on municipal bonds. Although we cannot directly measure institutional ownership, in Section 3.1, we discuss several indirect statistics to support the notion that taxable bonds attract more institutional ownership and trading compared to tax-exempt bonds.

could explain our findings. Second, we conduct a change analysis, using a matched sample, to examine disclosures around issuers' first taxable bond issuance and the call of their last outstanding taxable bond. Third, we exploit a tax law change that results in a plausibly exogenous increase in taxable bond issuances and examine disclosure changes around these exogenous taxable bond issuances.

Consistent with the issuance of taxable bonds shifting an issuer's clientele toward institutional investors who demand continuing disclosures, we find that issuers provide more continuing disclosures in years with taxable bonds outstanding. Our baseline estimate suggests that an issuer provides 19.5 percent more financial filings, on average, in years with outstanding taxable bonds, relative to years without. Using the mean number of 2.5 financial filings for the control group (i.e., issuer-years without taxable bonds) as the baseline, the estimated percentage increase implies 0.5 additional financial filings. When we examine material event notices (all continuing disclosures), we observe that the average issuer provides 32.7 (24.8) percent more disclosures in years with taxable bonds relative to years without.⁹ These results are consistent with tax clienteles in the municipal bond market shaping continuing disclosures.

To strengthen our identification strategy, we employ a stacked difference-in-differences design to examine changes in continuing disclosures following an issuer's (1) first taxable bond issuance and (2) call of its last outstanding taxable bond, relative to matched control samples. We expect issuers to provide more (fewer) continuing disclosures after they issue their first taxable bond (call their last taxable bond). Following an issuer's first taxable bond issuance, we observe a 12.5 percent increase in financial filings. Using the mean of 2.7 financial filings for the control group as the baseline, the percentage increase implies 0.3 additional financial filings. Issuers begin to increase disclosures in the year of the first taxable bond issuance, and the effect persists for up to three years afterward. Turning to the bond call analysis, following an issuer's call of its last taxable bond, we document a 5.4 percent decrease in financial filings, which corresponds to 0.1 fewer financial filings using the mean number of 2.7 filings for the control group as the baseline. The smaller estimated magnitude for the bond call analysis could be attributable to disclosures being "sticky" (e.g., [Einhorn and Ziv, 2008](#)). For both analyses, we do not observe significant pre-trends between treated and control issuers, which is consistent with the parallel trends assumption.

Lastly, we take advantage of a Tax Cuts and Jobs Act of 2017 (TCJA) provision that prohibits the use of tax-exempt bonds in advance refunding transactions, resulting in a plausibly exogenous increase in taxable bond issuances.¹⁰ We focus our analysis on a subsample of issuers that completed advance refunding transactions in both the pre- and post-TCJA periods, which allows us to hold the advance refunding decision constant and capture within-issuer changes in disclosure following an advance refunding transaction completed post-TCJA, relative to one completed pre-TCJA. Consistent with our primary analyses, we observe an increase in continuing disclosures following a post-TCJA advance refunding (i.e., where tax law mandates issuance of a taxable bond), as compared to the change in disclosures following a pre-TCJA advance refunding. Collectively, across multiple research specifications and settings, our results consistently suggest that municipal bond tax clienteles affect issuers' disclosure practices.

We conduct two additional analyses. First, we repeat our main analysis within the subsamples of issuers required to file continuing disclosures and those who are exempt from this requirement. Our main result holds in both subsamples, suggesting that the clientele for taxable bonds is associated with an increase in mandatory disclosures and an increase in voluntary disclosures among issuers that are exempt from continuing disclosure requirements. Second, given recent research that investors' reactions are stronger for timelier municipal disclosures ([Cuny et al., 2025](#)), we examine the timeliness of disclosures. We find some evidence that the timeliness of annual financial filings increases following an issuer's first taxable bond issuance.

Our study contributes to the literature on disclosure in the municipal bond market. While prior research examining disclosure noncompliance among municipal bond issuers largely focuses on the lack of regulation and enforcement (e.g., [Cuny et al., 2025](#); [Gillette et al., 2020](#); [Maffett et al., 2025](#)), we consider the taxability of municipal bonds and the corresponding tax clienteles in shaping issuers' disclosure practices. Our study also informs regulators of the municipal bond market, including the SEC and the MSRB, who seek to understand noncompliance with SEC Rule 15c2-12. Our results suggest that taxable municipal bonds, by attracting a tax clientele with stronger incentives and ability to demand and monitor disclosure, are associated with increased continuing disclosures. Our finding that investor demand shapes disclosure is particularly relevant in light of recent evidence that regulatory efforts can have a counterproductive effect on municipal disclosure compliance ([Maffett et al., 2025](#)). This insight broadens our understanding of why noncompliance persists in the municipal bond market and can be useful to regulators in improving the municipal disclosure environment.

2. Background, prior literature, and hypothesis development

2.1. Continuing disclosures in the municipal bond market

The municipal bond market is the largest capital market for state and local governments. As of 2022, there are approximately 50,000 issuers with \$4 trillion of municipal bonds outstanding ([SIFMA, 2023](#)). Unlike the disclosure environment for publicly traded firms, where an abundance of mandatory and voluntary disclosure is available, the disclosure environment for municipal bonds is

⁹ Using a baseline of 1.2 material event notices (3.3 continuing disclosures)—the control group mean—the estimated percentage increase implies an additional 0.4 material event notices (0.8 continuing disclosures).

¹⁰ Refunding transactions are used to refinance an existing bond. In a refunding transaction, the proceeds from a new bond (refunding bond) are used to pay off an existing bond (refunded bond). A current refunding occurs during the callable window of the refunded bond—usually 10 years after the original issuance—whereas an advance refunding occurs prior to the callable window (IRC Section 149(d)(5)). Section 4.5 provides additional details about this setting.

notoriously opaque. In this subsection, we discuss empirical evidence on the informativeness of municipal disclosures, disclosure regulation in the municipal market, and regulatory efforts to improve disclosure compliance.

The Governmental Accounting Standards Board (GASB) suggests that users of municipal financial reports include rating agency analysts, mutual funds, insurance companies, banks, underwriters, as well as taxpayers, government officials, and citizen advocacy groups.¹¹ Prior research finds that bond market participants use municipal financial reports to assess financial health, performance, and default risk (Gore, 2004; Ingram and Robbins, 1987).¹² The informativeness of municipal disclosures is also evidenced in the documented associations between municipal disclosures and credit ratings (e.g., Cuny et al., 2025), borrowing costs (e.g., Baber et al., 2013; Reck and Wilson, 2014), secondary market trading and pricing (Cuny et al., 2025; Reck and Wilson, 2006; Sherrill and Yerkes, 2018), and transaction costs (e.g., Cuny, 2018). Further, another stream of literature documents strategic financial reporting behavior among U.S. municipalities, which indirectly supports the notion that bond market participants use financial disclosures (e.g., Baber et al., 2024; Beck, 2018; Felix, 2014). Collectively, the evidence indicates that municipal issuers' continuing disclosures are informative for bond market participants.

Although municipal disclosures are informative, prior research suggests that many issuers fail to provide continuing disclosures. For example, Gillette et al. (2020) report that nearly 40 percent of issuers do not provide financial disclosures. Such noncompliance could be due to disclosure costs, including administrative costs, especially among smaller municipalities, and political costs, such as negative attention from the public (i.e., voters) or from federal and state regulators (e.g., Cuny, 2016; Gore et al., 2004; Ingram and DeJong, 1987). In addition, studies have attributed noncompliance with continuing disclosure requirements to the lack of direct regulatory oversight in the municipal bond market (Gillette et al., 2020; Maffett et al., 2025).

Tax-exempt bonds constitute the majority of outstanding municipal bonds; as such, the municipal market is dominated by retail investors. Importantly, even though retail investors may not review municipal disclosures directly, they may still benefit from better disclosures through, for example, more informed credit ratings (e.g., Cuny et al., 2025; Gore, 2004).¹³ Due to the large presence of retail investors, regulators have long been concerned about the lack of disclosure in this market. There are two primary regulators in the municipal bond market: the SEC and the MSRB. However, neither has direct authority over municipal bond issuers. To maintain state sovereignty, municipal bond issuers are largely exempt from federal regulations under the Securities and Exchange Acts of 1933 and 1934. The SEC indirectly regulates municipal disclosures by requiring underwriters to obtain a commitment from issuers to file continuing disclosures, but underwriters cannot directly enforce continuing disclosure requirements after the bond offering (Maffett et al., 2025). While underwriters can refuse to underwrite future offerings if an issuer fails to comply with continuing disclosure requirements, doing so might impose reputational costs. Similarly, the MSRB can regulate municipal securities professionals (e.g., dealers and advisors) but has no direct authority over municipal bond issuers (MSRB, 2022a).

In 2014, as part of its ongoing efforts to enhance municipal disclosure, the SEC launched the Municipalities Continuing Disclosure Cooperation (MCDC) initiative, a regulatory amnesty program that granted favorable settlement terms to issuers and underwriters who voluntarily self-reported prior disclosure noncompliance. Maffett et al. (2025) find that the MCDC initiative resulted in an approximately 10 percent decrease in overall continuing disclosure compliance. The failure of the SEC's most recent effort to improve municipal disclosures suggests that, in addition to regulatory enforcement, it may be necessary to consider the role of market forces, such as investor demand and monitoring, in shaping municipalities' continuing disclosures.

2.2. Tax status and clientele for municipal bonds

Although most municipal bonds are tax-exempt, some are taxable (i.e., interest income is subject to federal income tax). Holding risk constant, issuing taxable municipal bonds is costlier for municipalities than issuing tax-exempt bonds because investors demand a higher pretax rate of return to compensate for the tax costs on the interest income. Despite the higher borrowing costs, municipalities issue taxable municipal bonds for three main reasons. First, taxable bonds provide the issuer with greater flexibility in the use of proceeds. To qualify for tax exemption, issuers must meet certain requirements under the Internal Revenue Code. For example, generally no more than 10 percent of the proceeds can be used for any private business purpose (IRC Section 141(b)). If a bond does not meet this requirement, it will be classified as a private activity bond. Private activity bonds may still qualify for tax exemption. However, there are limits on the total dollar value of tax-exempt private activity bonds a state can issue each year.¹⁴ Issuers can circumvent these limits by simply issuing taxable bonds.

¹¹ <https://www.gasb.org/about-us/gasb-and-the-user-community/financial-statement-users> (last accessed April 2025).

¹² Despite the low realized default rates, bond prices are highly sensitive to small changes in risk, and research estimates that 74 to 84 percent of bond yields are attributable to default risk (Schwert, 2017).

¹³ Credit rating agencies use a mix of inputs to assess credit risk. For example, Moody's adopts a scorecard approach for U.S. cities and counties, which primarily considers economic factors (e.g., local resident income), financial performance (e.g., available fund balance), institutional framework (e.g., issuers' ability to impose taxes), and leverage. Moody's disclosed methodology explicitly cites audited financial statements as the data source for assessing financial performance and leverage, and considers the lack of financial disclosures as an indicator for downward credit rating adjustments (Moody's, 2024). In addition to public disclosures, rating agencies may obtain private information through site visits or directly request non-public documents (Government Finance Officers Association, 2022).

¹⁴ For example, in 2021, each state could only issue tax-exempt private activity bonds up to the greater of \$110 per capita or \$325 million (Hernandez Barcena and Wessel, 2020). Additionally, interest income from tax-exempt private activity bonds is generally included in gross income for purposes of determining the Alternative Minimum Tax (AMT), with certain exceptions for qualified bonds issued by municipalities following a disaster.

Second, by issuing taxable bonds, municipalities can broaden their investor base. Investors such as pension funds and university endowments that generally do not pay federal income taxes, as well as mutual funds and ETFs, which are typically taxed as flow-through entities, may not value the tax-exempt feature of municipal bonds. By issuing taxable bonds, which provide higher pretax returns, issuers can attract investors who do not benefit from tax exemption. For example, to help state and local governments raise funds, the American Recovery and Reinvestment Act of 2009 created Build America Bonds (BABs). Interest income from BABs is fully taxable to investors, and municipalities receive reimbursements from the federal government to compensate for the higher pretax cost of issuing a taxable bond. BABs led to a wave of taxable municipal bond issuances, with over \$151 billion in taxable bonds issued in 2010 (Tilley et al., 2021). Consistent with taxable bonds attracting a broader investor base, institutional ownership of municipal bonds increased with the BABs program (Bagley et al., 2022).

Third, the TCJA introduced a provision that requires the issuance of taxable bonds in advance refunding transactions. Koutney et al. (2023) find a large spike in advance refunding transactions in December 2017, before the TCJA went into effect, suggesting that municipalities are responsive to expected increases in pretax borrowing costs. This TCJA provision, combined with low interest rates in 2020 and 2021, led to a second wave of taxable municipal bond issuances. In the first eleven months of 2020, \$129.3 billion in taxable bonds were issued, nearly twice (five times) the amount issued in 2019 (2018) (Hernandez Barcena and Wessel, 2020).

Because interest income from most municipal bonds is exempt from federal taxation, individuals in higher income tax brackets form a primary tax clientele for tax-exempt municipal bonds. Due to the tax-exemption benefit, these high-tax investors are willing to bear an implicit tax, i.e., a lower pretax return on such bonds (Erickson et al., 2020). Consistent with high-tax retail investors forming a primary clientele for tax-exempt bonds, prior research documents that bond pricing is sensitive to individual tax rates (e.g., Ang et al., 2010; Green, 1993; Schaefer, 1982). While tax-exempt bonds largely attract high-tax retail investors, we expect a broader set of institutional investors to invest in taxable municipal bonds (Schwert, 2017).¹⁵ These institutional investors form a natural clientele for taxable municipal bonds that provide a higher pretax yield.¹⁶

2.3. Hypothesis development

Prior research provides evidence that institutional investors serve a key monitoring role in equity markets (e.g., Bebchuk et al., 2017; Hartzell and Starks, 2003; Shleifer and Vishny, 1986). Bushee and Noe (2000) suggest that improved disclosures reduce institutional investors' monitoring costs. Bird and Karolyi (2016) find that firms increase the quantity and quality of disclosures after positive shocks to institutional ownership, and Abramova et al. (2020) document a positive relation between institutional investor attention and the quantity of voluntary disclosures. Collectively, these studies suggest that institutional investors demand disclosures and firms respond to this demand in the equity market.

As discussed in Section 2.1, municipal continuing disclosures are useful for bond market participants to assess financial health, performance, and default risk. Although trading volume is lower in the municipal bond market than in the equity market, institutional investors monitor and price their positions frequently. Thus, they have incentives to continuously search for relevant economic and fundamental information about issuers and bonds (Cuny, 2018). For example, the Head of Municipal Bond Credit Research at an asset management company notes, "As an institutional investor, my team of analysts and I rely heavily on timely, accurate, and relevant financial disclosure to help our portfolio managers steward some \$18 billion of shareholder investments. A detailed review of issuers' notes to the financial statements is a key component of any robust credit analysis."¹⁷ Compared to institutional investors, dispersed retail investors are unlikely to exert direct influence on issuers' disclosure practices. Retail investors may also be less attentive to bond market news, including continuing disclosures (Cornaggia et al., 2022). Further, retail investors commonly hold bonds to maturity and rely on credit ratings, suggesting that they may have weaker incentives to actively demand disclosures. Hence, as taxable bonds shift the tax clientele from retail investors toward institutional investors, who are able and have incentives to demand more disclosures, we expect issuers with outstanding taxable municipal bonds to provide more continuing disclosures. We state our hypothesis in the alternative form:

H1. The clientele shift related to taxable municipal bonds is positively associated with municipal bond issuers' continuing disclosures.

Yet, we may not observe differences in continuing disclosures based on issuers' tax clienteles. First, Cuny et al. (2025) find evidence of trading before disclosure filings. Their finding, along with the fact that Regulation Fair Disclosure (Reg FD) does not apply to municipal issuers, suggests that institutional investors may obtain information via other means. If this is the case, institutional investors may not demand public municipal disclosures. Second, the municipal bond market differs from the equity market in that

¹⁵ Consistent with the noted preferences, individual investors directly own the plurality of municipal bonds, even though institutional investors also invest in municipal bonds. Bagley et al. (2022, p.9) find that "fund holdings of municipal securities (including mutual funds, ETFs and CEFs)" have increased significantly in recent years. Yet, to the extent that taxable bonds attract more institutional investors than tax-exempt bonds, we expect a shift in clientele when municipalities issue taxable bonds.

¹⁶ When comparing bond yields at issuance, we find that the average offering yield for taxable bonds is consistently higher than that for tax-exempt bonds. We estimate implicit tax rates between 15 and 30 percent in most years, which are below the top individual marginal tax rate. While we caveat that the implicit tax rate calculation assumes that tax-exempt and taxable bonds are of similar risk, finding consistently higher yields for taxable bonds validates the existence of implicit taxes—an important driver of the tax clientele effect.

¹⁷ https://www.gasb.org/page/ShowPdf?path=DISC-Revised_ED_CL033.pdf, last accessed April 2025.

municipal disclosures are not as timely, municipal bonds are not frequently traded in the secondary market, and the default risk has historically been low (DPC DATA, 2024; MSRB, 2022b). Thus, findings based on the equity market might not apply to the municipal bond market. Third, while high-tax retail investors form a primary clientele for tax-exempt bonds, institutional investors such as mutual funds or ETFs also hold and trade tax-exempt securities. Using macro-level data, Bagley et al. (2022) document a gradual shift among retail investors from direct ownership toward indirect ownership via mutual funds and ETFs. While individual direct ownership is still the dominant means of municipal bond investment, the trend toward indirect ownership could lead to an increase in institutional investors' demand for disclosure across all issuers, including those with only tax-exempt bonds. Thus, tax clienteles may not affect issuers' continuing disclosures.

3. Research design and sample selection

3.1. Research design

To examine the relation between municipal bond tax clienteles and issuers' disclosure practices, we begin by estimating the following model:

$$\text{Disclosure} = \beta_1 + \beta_2 \text{Ind. Taxable Bond} + \Sigma\beta \text{Controls} + \text{Issuer Fixed Effects} + \text{Year Fixed Effects} + \varepsilon \quad (1)$$

We use three measures to capture an issuer's continuing disclosures (*Disclosure*). Our primary measure, *#Financial Filings*, captures the number of financial filings, including annual and quarterly financial statements, budgets, and operating data, filed during the fiscal year.¹⁸ Additionally, we measure the number of material event notices filed during the fiscal year (*#Event Filings*) and all continuing disclosures made by an issuer in a fiscal year (*#Continuing Disclosures*).¹⁹ Given that our dependent variables are count variables, we use a fixed-effects Poisson model to estimate equation (1) (Cohn et al., 2022). We use the taxability of municipal bonds to capture the underlying construct of tax clienteles. *Ind. Taxable Bond* is an indicator variable set to one if an issuer has at least one taxable bond outstanding at the end of the fiscal year, and zero otherwise. Although institutional investors can hold both tax-exempt and taxable bonds, taxable bonds are more likely to attract institutional investors. Thus, having a taxable bond will shift an issuer's clientele from individuals to institutions.²⁰

We include issuer fixed effects to control for time-invariant issuer characteristics. This research design is important because it provides within-issuer estimates, mitigating concerns that issuers with taxable bonds may be fundamentally different from issuers without. Accordingly, β_2 in equation (1) captures the difference in an issuer's continuing disclosures during years with taxable bonds outstanding relative to years without. We also include year fixed effects to control for broader economic conditions and to capture any common trends in disclosures over time.

Equation (1) includes control variables that prior research suggests are correlated with an issuer's continuing disclosure practices. First, we control for the number of bonds outstanding ($\text{Log}(\text{Bond Count})$) and the amount of new bonds issued during the fiscal year ($\text{Log}(\text{New Issue Amt})$) to account for the possibility that issuers with more outstanding or newly issued debt provide more continuing disclosures (e.g., Gillette et al., 2020). Second, prior research finds that callable bonds are more likely to be used when information asymmetry is high (e.g., Banko and Zhou, 2010), and general obligation (GO) bonds are considered less risky because they are backed by the full faith and credit of the issuer. Thus, we control for the percentage of an issuer's outstanding bonds that are callable bonds (*Callable Bond%*) and GO bonds (*GO Bond%*). Third, following Maffett et al. (2025), we control for issuers' incentives to refinance and short-term liquidity needs using the total principal amount of debt maturing in the current fiscal year ($\text{Log}(\text{Amt Due})$) and the shortest maturity (measured in years) among bonds outstanding at the end of the fiscal year (*Maturity*).

Fourth, we control for the issuer's average credit rating across all outstanding bonds (*Credit Rating*), given evidence in Gillette et al. (2020) that issuers reduce disclosures when credit ratings are higher. Controlling for credit ratings also alleviates the concern that taxable bonds may have different risk profiles, which could influence disclosure. Credit ratings are not well-populated; thus, to preserve the sample size, we include an indicator variable set to one when credit ratings are missing (*Credit Rating Missing*) and reset missing *Credit Rating* values to zero. Last, we include changes in unemployment rate ($\Delta\text{Unemployment}$) and state GDP (ΔGDP) to control for time-varying state-level economic conditions. We cluster standard errors at the issuer level. All variables are defined in Appendix A.

To provide a stronger identification, we also test H1 using the following fixed-effects Poisson regression to examine changes in

¹⁸ Our main results are qualitatively similar when we examine annual financial statements only.

¹⁹ Material event notices include, in brief, notices of delinquencies, defaults, acceleration or termination events, unscheduled draws, adverse tax opinions, modifications to the rights of security holders or terms, material bond calls, defeasances, ratings changes, bankruptcy, insolvency, merger or acquisition activity, or changes in trustees. These filings are triggered by material events, similar to Form 8-K filings for publicly traded companies.

²⁰ It is empirically challenging to measure municipal bond ownership; thus, we provide two indirect statistics to support the notion that taxable municipal bonds are more likely to attract institutional investors. First, the average trade size (skewness in trade size) for tax-exempt bonds is \$256,412 (579.92), as compared to \$519,622 (145.36) for taxable bonds (MSRB, 2022b). This suggests that tax-exempt (taxable) bonds have smaller (larger) trade sizes and a greater number of (fewer) small trades, which is indicative of more institutional trades for taxable bonds. Second, among the issuer-years with data in CRSP to identify mutual fund ownership (38 percent of our primary sample), we find that aggregate mutual fund holdings—a specific type of institutional ownership—are on average 2.18 times larger for issuer-years with taxable bonds outstanding compared to issuer-years without. Overall, these statistics are consistent with taxable bonds shifting an issuer's investor clientele toward institutional investors.

continuing disclosures around an issuer's *first* taxable bond issuance:

$$\text{Disclosure} = \beta_1 + \beta_2 \text{ Issue Taxable Bond} * \text{Post} + \Sigma \beta \text{ Controls} + \text{Issuer-Cohort Fixed Effects} + \text{Relative Year} - \text{Cohort Fixed Effects} + \varepsilon \quad (2)$$

Issue Taxable Bond is an indicator set to one for issuers that issued a taxable bond for the first time during our sample period (i.e., treated issuers), and zero otherwise. We identify an issuer's first taxable bond issuance using two criteria: (1) the issuer has a new taxable bond issuance during the year, and (2) the issuer had no taxable bonds outstanding at the end of the prior year.²¹ We estimate equation (2) over an eight-year window from $t-4$ to $t+3$, where year t is the year of the first taxable bond issuance (i.e., the “event year”). *Post* is set equal to one for years t through $t+3$, and zero otherwise. We use issuers that have never issued a taxable bond to construct the control group. To ensure comparability, we match each treated issuer with up to five control issuers, with replacement, that are in the same state, issue bonds for the same purpose, and are closest in size—with a maximum size difference of 10 percent—based on the amount of total outstanding municipal debt at the end of year t .²² We require issuers to have at least one year of non-missing data in both the pre- and post-periods.

Because event years vary by issuer, we implement a stacked difference-in-differences design and define cohorts based on the event year (Baker et al., 2022). With the use of a stacked, matched sample, equation (2) includes issuer-cohort and relative year-cohort fixed effects. The inclusion of issuer-cohort fixed effects aligns first taxable bond issuances by event time, such that the identification captures changes in disclosure over time for the same issuer for a given taxable bond issuance. Due to this fixed effect structure, the main effects of *Issue Taxable Bond* and *Post* are subsumed. The coefficient of interest is the interaction term, *Issue Taxable Bond* * *Post* (β_2), which captures the incremental change in continuing disclosures following an issuer's first taxable bond issuance, relative to control issuers. Because equation (2) focuses on new issuances, we replace *Callable Bond%* (*GO Bond%*) with *New Callable Bond%* (*New GO Bond%*) to capture the percentage of new issuances that are callable (GO) bonds. All other control variables and dependent variables are as previously defined.

We also examine changes in disclosures after an issuer calls its last taxable bond using the following fixed-effects Poisson regression:

$$\text{Disclosure} = \beta_1 + \beta_2 \text{ Call Taxable Bond} * \text{Post} + \Sigma \beta \text{ Controls} + \text{Issuer-Cohort Fixed Effects} + \text{Relative Year} - \text{Cohort Fixed Effects} + \varepsilon \quad (3)$$

Call Taxable Bond is an indicator set equal to one if an issuer calls its last outstanding taxable bond, and zero otherwise. We identify the call of an issuer's last taxable bond using three criteria: (1) the issuer had a taxable bond outstanding at the end of the prior year, (2) the issuer does not have a taxable bond outstanding at the end of the current year, and (3) the issuer does not issue a new taxable bond during the remainder of our sample period. The year in which an issuer calls its last taxable bond is year t (i.e., the “event year”). Our approach to identifying a matched control sample is the same as the approach used in the first taxable bond issuance analysis. The control variables and empirical specification for equation (3) also follow those in equation (2). The coefficient of interest is the interaction term, *Call Taxable Bond* * *Post* (β_2), which captures the incremental change in continuing disclosures after an issuer calls its last outstanding taxable bond, relative to the change in disclosures for the matched control issuers.

3.2. Sample selection

To construct our sample, we identify all unique issuers from the Mergent Municipal Bond Securities Database (Mergent) with fiscal years beginning on or after July 1, 2009 and ending on or before December 31, 2022. Our sample period begins on July 1, 2009, when the MSRB introduced Electronic Municipal Market Access (EMMA), a centralized repository for municipal disclosures, and ends on December 31, 2022, the latest date for which disclosure data were available at the time of data collection. Thus, the disclosure dissemination channel is held constant over our sample period (Cuny, 2016). We exclude bonds with a zero outstanding balance at the end of the fiscal year. We also exclude U.S. territories, restricting the sample to issuers located in the 50 U.S. states and Washington, D. C. For each issuer, we obtain bond characteristics from Mergent to construct our measures of taxable bonds as well as control variables. We aggregate bond-specific variables to the issuer-year level, which is the unit of analysis for our regressions.

We obtain information about issuers' continuing disclosures from the MSRB. For a given issuer-year, we match the disclosure to the Mergent base sample using the issuer CUSIP and the filing date. For example, we match a disclosure filed on May 15, 2018 to an issuer's fiscal year encompassing this filing date. We obtain the fiscal year-end date for each issuer based on its annual financial filings. Consistent with prior research, we use a June 30 fiscal year for issuers missing fiscal year-end information (e.g., Gillette et al., 2020).

²¹ We acknowledge that our approach could capture issuers that previously issued and called a taxable bond before the beginning of our sample period. This is a limitation given the data available. To the extent that the shift in tax clientele is most salient around an issuer's first taxable bond issuance, including issuers with outstanding taxable bonds prior to our sample period should work against us finding a significant tax clientele effect on disclosure.

²² We categorize bond purposes into 10 groups (see Table 2, Panel B and Section 4.1 for details). In our sample, 84.6 percent of issuer-years have outstanding bonds in only one bond purpose group, and 12.3 (3.1) percent have outstanding bonds in two (more than two) purpose groups. When matching treated and control issuers, we retain issuers with outstanding bonds in up to two purpose groups to maximize sample size without compromising the quality of matching. For issuers with two bond purposes, we require that the matched control issuers have outstanding bonds in the exact same two purpose groups.

For issuer-years without a matched disclosure from the MSRB, we reset the corresponding disclosure variables to zero.²³ After requiring non-missing data to compute regression variables, the sample used to estimate equation (1) includes 560,652 issuer-years and 59,256 unique issuers. Table 1 presents sample selection procedures. We also report the number of observations used to estimate equations (2) and (3) at the issuer-year and issuer level in Table 1.

4. Results

4.1. Descriptive statistics

Fig. 1 plots the average number of total bond issuances, tax-exempt bond issuances, and taxable bond issuances per issuer, by fiscal year. The majority of bond issuances, ranging between 82.0 and 94.0 percent of the total number of new issuances each year (untabulated), are tax-exempt bonds. The average number of taxable bond issuances was relatively stable in the middle portion of our sample period, ranging between 0.11 and 0.21 from 2011 to 2019. We observe a higher number of taxable bond issuances in 2010, with a per issuer average of 0.34, which coincides with the BABs program. We observe another increase after 2019, with a per issuer average of 0.36 (representing 14.0 percent of new issuances) in 2020 and 0.51 (representing 18.0 percent of new issuances) in 2021. This increasing trend is likely driven by both the TCJA provision that prohibited the use of tax-exempt bonds for advance refunding transactions and historically low interest rates. Overall, the time trend of taxable bond issuances in our sample is largely consistent with practitioner publications (Hernandez Barcena and Wessel, 2020).

Table 2, Panel A presents descriptive statistics for the primary sample. On average, issuers provide 2.2 financial filings (3.5 continuing disclosures) per year. We observe that 17.9 percent of issuer-years have at least one taxable bond outstanding. On average, callable bonds and GO bonds represent 69.1 and 39.9 percent of the total dollar value of outstanding bonds, respectively.²⁴ Table 2, Panel B presents descriptive statistics related to the use of bond proceeds (i.e., bond purpose). Our sample issuers report 61 unique bond purpose descriptions, which we categorize into ten groups. For most issuer-years (84.6 percent), all bonds outstanding are associated with one bond purpose group. When examining the distribution of bond purpose conditional on whether an issuer-year has a taxable bond outstanding (i.e., *Ind. Taxable Bond* = 1), we find that issuer-years with a taxable bond outstanding are less likely to have bonds that fund Public Services or Utilities and more likely to have multiple bond purposes.²⁵

4.2. Primary results

Fig. 2 plots the average number of financial filings for issuer-years with at least one taxable bond outstanding and those without. We observe an increase in financial filings over the sample period for issuer-years both with and without taxable bonds. On average, issuer-years with taxable bonds consistently provide more financial filings relative to issuer-years without taxable bonds, which provides preliminary support for H1.

Table 3 presents the results of estimating equation (1).²⁶ If taxable bonds attract a clientele that demands additional continuing disclosures, we should observe a positive and significant coefficient on *Ind. Taxable Bond* (i.e., $\beta_2 > 0$). In column 1, where the dependent variable is *#Financial Filings*, we observe a positive and significant coefficient on *Ind. Taxable Bond*. The coefficient estimate suggests that in years when an issuer has a taxable bond outstanding, the average number of financial filings is 19.5 percent higher (i.e., $\exp(0.1780)-1$) relative to when the issuer does not have a taxable bond outstanding.²⁷ Using the mean number of 2.5 financial filings for the control group as the baseline (untabulated), the estimated percentage increase implies 0.5 additional financial filings.

The coefficient estimates on the control variables are largely consistent with our expectations and prior research. In column 1, we observe more *#Financial Filings* in years when the issuer has more bonds outstanding (*Log(Bond Count)*), new issuances (*Log(New Issue Amt)*), or a larger amount of debt maturing within the year (*Log(Amt Due)*). Consistent with Gillette et al. (2020), we observe fewer *#Financial Filings* in years with higher credit ratings. If disclosure is generally increasing in risk, we expect *#Financial Filings* to be

²³ We test the robustness of our results to these research design choices using an alternative sample that only includes issuers that provide at least one disclosure to the MSRB over the sample period and issuers with a fiscal year-end date based on MSRB filings. We find qualitatively similar results. See Section 5.2 and the Online Appendix.

²⁴ The designation as a callable bond or a GO bond is not mutually exclusive.

²⁵ We find that an issuer's bond purpose is "sticky" over time. Within the subsample of issuer-years with one bond purpose group, 97 percent of issuers report the same bond purpose over our sample period. Thus, we cannot include bond purposes and issuer fixed effects. In an untabulated analysis, we re-estimate equation (1) replacing issuer fixed effects with bond purpose indicators and find similar results. We do not repeat this analysis for the first taxable bond issuance test and the call of last taxable bond test because we match treated and control issuers on bond purpose.

²⁶ As noted in Cohn et al. (2022), the fixed-effects Poisson regression excludes a group if the outcome variable is zero for all observations in the group. Thus, each table notes the number of observations used in the Poisson estimation.

²⁷ To provide a reference point to evaluate the economic magnitude, Maffett et al. (2025) find that issuers decrease continuing disclosures by 10 percent after the SEC's MCDC initiative compared to their pre-MCDC initiative levels. In another study, Gillette et al. (2020) find a 5.1 percent decrease in the frequency of continuing financial disclosures following Moody's recalibration of the municipal ratings scale, which decreased the market's demand for continuing disclosures. A third study, Cheng et al. (2023), finds that issuers who are disadvantaged by one notch in the credit rating (i.e., receive fewer credit rating upgrades due to Moody's recalibration in 2010) increase their annual financial disclosures by 40 percent. Taken together, these comparisons suggest that our estimate is economically meaningful and broadly consistent with the magnitudes reported in prior research.

Table 1
Sample selection.

	Issuer-Year Observations	Issuer Observations
U.S. municipal bond issuers in Mergent, with outstanding bonds, with fiscal years beginning on or after July 1, 2009 and ending on or before December 31, 2022	561,775	59,360
Less: observations for issuers in U.S. territories (i.e., AS, GU, MP, NA, PR, or VI)	(1,123)	(104)
Primary Sample (equation (1))	560,652	59,256
Observations with a first taxable bond issuance during our sample period (<i>Issue Taxable Bond</i> = 1)	23,586	3,651
Matched Control Observations	87,602	13,766
First Taxable Bond Issuance Sample (equation (2))	111,188	17,417
Observations with a taxable bond call during our sample period (<i>Call Taxable Bond</i> = 1)	10,168	1,517
Matched Control Observations	37,624	5,699
Taxable Bond Call Sample (equation (3))	47,792	7,216

This table reports the sample construction for the samples used to examine the relation between the tax clienteles and the disclosure practices of municipal bond issuers (H1).

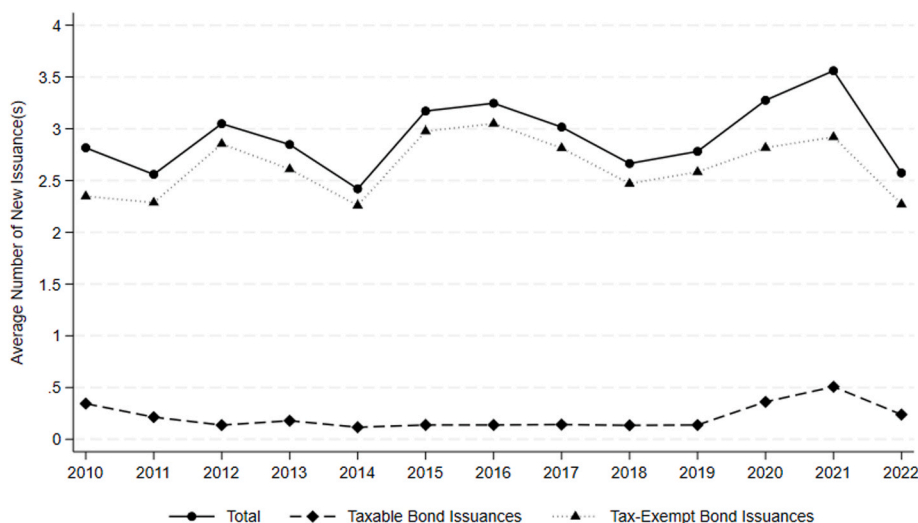


Fig. 1. The Average Number of Bond Issuances per Issuer, by Fiscal Year and Issue Type

Notes: Fig. 1 plots the average number of total bond issuances, tax-exempt bond issuances, and taxable bond issuances per issuer, by fiscal year.

positively associated with callable bonds (*Callable Bond%*) and negatively associated with GO bonds (*GO Bond%*). However, our results are inconsistent with this expectation. Prior research generally finds an insignificant relation between disclosure and *Callable Bond%* and *GO Bond%* (Gillette et al., 2020; Maffett et al., 2025). Thus, there is a lack of consistent evidence on the relation between disclosure and *Callable Bond%* and *GO Bond%*. We also observe fewer *#Financial Filings* in years with increasing GDP, consistent with evidence in Cuny (2016) that credit risk is lower during economic growth, resulting in fewer disclosures. At odds with the coefficient on ΔGDP is the negative relation between *#Financial Filings* and the change in unemployment ($\Delta Unemployment$). However, this result is consistent with Cheng et al. (2023). A possible explanation is that local officials' electoral incentives or concerns about reputational damage inhibit disclosure when the unemployment rate increases (Cuny, 2016).

In columns 2 and 3, where the dependent variables are *#Event Filings* and *#Continuing Disclosures*, respectively, we also observe a positive and significant coefficient on *Ind. Taxable Bond*. Our estimates indicate that issuers provide 32.7 (24.8) percent more event filings (continuing disclosures) in years with taxable bonds relative to years without, or 0.4 (0.8) additional filings using the mean of 1.2 event filings (3.3 continuing disclosures) for the control group as the baseline (untabulated). Collectively, the findings in Table 3 suggest a positive and significant relation between having a taxable bond outstanding and municipal bond issuers' disclosure levels. This evidence is consistent with the clientele for taxable bonds demanding additional disclosures and issuers responding to such demand.

4.3. Examination of first taxable bond issuances

Given the "stickiness" of disclosure and the long maturity of municipal bonds, if issuers change their level of continuing disclosure in response to investor demand, we expect the tax clientele effect on disclosure to be most salient around the first issuance of a taxable bond. We first provide univariate evidence in Fig. 3, Panel A by plotting the number of financial filings around the first taxable bond

Table 2
Descriptive statistics.

Panel A: Descriptive Statistics for the Primary Sample (N = 560,652)				
	Mean	St. Dev.	Median	
#Financial Filings	2.175	3.429	1.000	
#Event Filings	1.213	2.768	0.000	
#Continuing Disclosures	3.464	5.684	2.000	
Ind. Taxable Bond	0.179	0.383	0.000	
Log(Bond Count)	2.475	1.215	2.485	
Log(New Issue Amount)	3.364	6.561	0.000	
Callable Bond%	0.691	0.323	0.785	
GO Bond%	0.399	0.470	0.000	
Log(Amt Due)	10.123	6.156	12.722	
Maturity	2.349	4.545	1.000	
Credit Rating	12.727	8.183	17.000	
Credit Rating Missing	0.281	0.450	0.000	
Δ Unemployment	-0.009	0.475	-0.098	
Δ GDP	0.047	0.048	0.042	
Panel B: Sample by Bond Purpose Grouping				
	Example Bond Purpose Description (before categorization)	Primary (full) Sample	Ind. Taxable Bond = 1	Ind. Taxable Bond = 0
(1)	Agriculture			
	Irrigation	392	0.03%	0.08%
(2)	Economic and Industrial Development			
	Industrial development	24,639	5.60%	4.13%
(3)	Education			
	Primary/Secondary education	155,814	28.71%	27.59%
(4)	Healthcare			
	Hospitals	27,921	3.20%	5.37%
(5)	Public Housing			
	Multi-family housing	26,366	5.16%	4.60%
(6)	Public Recreation			
	Parks/Zoos/Beaches	10,707	0.71%	2.17%
(7)	Public Services			
	Courts	135,018	17.74%	25.47%
(8)	Transportation and Infrastructure			
	Toll road and highway	8944	1.68%	1.58%
(9)	Utilities			
	Public power	83,716	7.81%	16.48%
(10)	Unclassified			
	N/A	792	0.13%	0.14%
(11)	Multiple			
	N/A	86,343	29.22%	12.39%
	TOTAL	560,652	100%	100%

Panel A provides descriptive statistics for the variables used in estimating equation (1), for the full sample. Variables are defined in Appendix A.

Panel B reports the sample by bond purpose grouping. Our sample issuers report 61 unique bond purpose descriptions, which we categorize into ten groups.

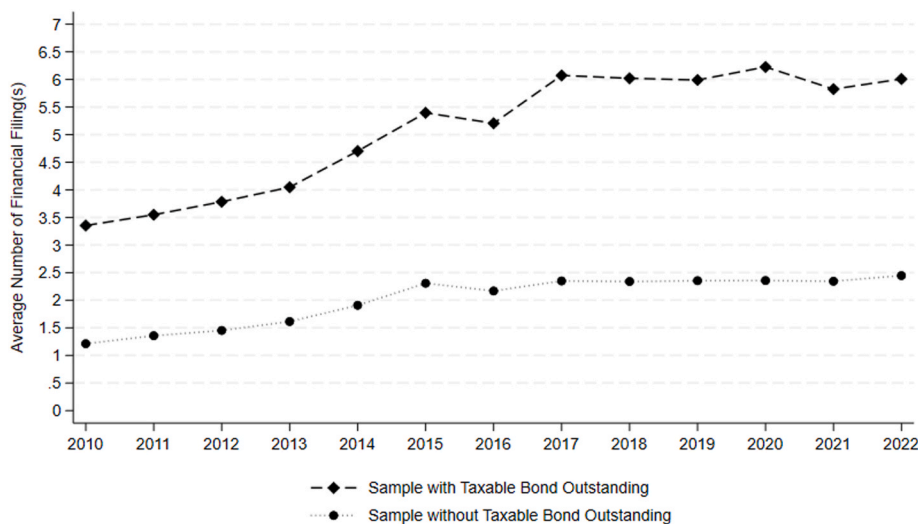


Fig. 2. The Average Number of Financial Filings per Issuer, by Fiscal Year for the Subsamples of Issuer-Years with and without Outstanding Taxable Bonds

Notes: Fig. 2 plots the average number of financial filings per issuer, by fiscal year, for the subsample of observations with a taxable bond outstanding in a given year (*Ind. Taxable Bond* = 1; N = 100,340), and the subsample of observations without a taxable bond outstanding (N = 460,312).

Table 3

The effect of taxable bonds on the quantity of continuing disclosures.

DV =	Pred. Sign	(1)	(2)	(3)
		Coeff.	Coeff.	Coeff.
		(z-stat)	(z-stat)	(z-stat)
<i>Ind. Taxable Bond</i>	+	<i>#Financial Filings</i> 0.1780*** (8.67)	<i>#Event Filings</i> 0.2830*** (11.79)	<i>#Continuing Disclosures</i> 0.2214*** (12.04)
<i>Log(Bond Count)</i>		0.3608*** (31.51)	0.4703*** (29.98)	0.3995*** (36.59)
<i>Log(New Issue Amt)</i>		0.0013*** (2.71)	0.0182*** (22.25)	0.0076*** (15.51)
<i>Callable Bond%</i>		-0.0817*** (-3.79)	-0.2455*** (-6.24)	-0.1251*** (-5.44)
<i>GO Bond%</i>		0.0950*** (4.40)	0.1937*** (5.74)	0.1567*** (7.45)
<i>Log(Amt Due)</i>		0.0186*** (11.99)	0.0074*** (4.70)	0.0150*** (11.06)
<i>Maturity</i>		0.0146*** (5.89)	0.0195*** (7.09)	0.0179*** (8.40)
<i>Credit Rating</i>		-0.4926*** (-4.82)	-0.9061*** (-6.20)	-0.6302*** (-6.39)
<i>Credit Rating Missing</i>		-0.0233*** (-4.02)	-0.0474*** (-5.70)	-0.0307*** (-5.51)
<i>ΔUnemployment</i>		-0.0200*** (-2.75)	-0.0997*** (-7.47)	-0.0443*** (-6.01)
<i>ΔGDP</i>		-0.4819*** (-5.64)	-0.5364*** (-3.59)	-0.4814*** (-5.48)
<i>Fixed Effects</i>		Issuer and Year	Issuer and Year	Issuer and Year
<i># of Observations</i>		560,652	560,652	560,652
<i># of Obs. Used in Poisson Estimation</i>		452,448	463,622	502,589
<i>Pseudo R²</i>		0.552	0.590	0.627

This table presents the results of a Poisson regression estimating equation (1). In column 1, the dependent variable is *#Financial Filings*, defined as the number of financial filings made by an issuer during the fiscal year. In columns 2 and 3, the dependent variables are the number of event filings (*#Event Filings*) and the total number of all continuing disclosures (*#Continuing Disclosures*) made by an issuer in a fiscal year, respectively. *Ind. Taxable Bond* is an indicator variable equal to one if the issuer has at least one taxable bond outstanding at the end of the year. We expect the coefficient on *Ind. Taxable Bond* to be positive, as indicated in the brackets. We include issuer fixed effects and year fixed effects. Standard errors are clustered at the issuer level. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively, using a one-tailed t-test where we denote a prediction, and a two-tailed t-test otherwise. Variables are defined in [Appendix A](#).

issuance. We define the event year (year t) as the year of the treated issuer's first taxable bond issuance. Treated and control issuers have similar disclosure levels before year t , suggesting that our matching procedure is effective in identifying comparable control issuers. We observe an increase in disclosure beginning in the year of the first taxable bond issuance and persisting through $t+3$, whereas we do not observe a similar increase for control issuers. Treated (control) issuers provide 2.46 (2.48) financial filings on average during years $t-4$ to $t-1$ and provide 3.20 (2.49) financial filings during years t to $t+3$ (untabulated). Overall, this figure suggests that an issuer's first taxable bond issuance is associated with an increase in financial filings.

[Table 4](#) presents the results from estimating a stacked difference-in-differences model to examine changes in continuing disclosures following an issuer's first taxable bond issuance (equation (2)). Across all columns, we observe a positive and significant coefficient on *Issue Taxable Bond*Post*. The results are consistent with the univariate patterns in [Fig. 3](#), Panel A and suggest that treated issuers increase continuing disclosures following their first taxable bond issuance, relative to matched control issuers. In column 1, the coefficient estimate of 0.1178 suggests that following treated issuers' first taxable bond issuance, the number of financial filings increases by 12.5 percent. Using the mean number of 2.7 financial filings for the control issuers in the pre-period as the baseline (untabulated), the estimated percentage increase implies 0.3 additional filings. For event filings (continuing disclosures), we estimate an increase in disclosure of 29.5 (17.5) percent, or 0.4 (0.7) additional filings using the mean number of 1.3 event filings (3.6 continuing disclosures) for the control issuers in the pre-period as the baseline (untabulated).²⁸ Overall, [Table 4](#) provides further evidence on the positive relation between an issuer's tax clientele and disclosure practices, supporting [H1](#).

An assumption underlying the difference-in-differences design is that the disclosures by the treated and control issuers would have evolved in a comparable manner, absent the issuance of a taxable bond (i.e., the parallel trends assumption). [Fig. 3](#), Panel A provides

²⁸ We note that 27.7 percent of treated issuers also issue subsequent taxable bond(s). However, re-estimating equation (2) to quantify the incremental effect of subsequent issuances (if any) is challenging because 60.4 percent of the second issuances occur within three years after the first issuances, creating substantial overlap in the post-issuance window for the first issuance and the pre-issuance window for the second issuance. We confirm that [Table 4](#) results are robust if we restrict the analysis to the subsample of treated issuers with only one taxable bond issuance (untabulated).

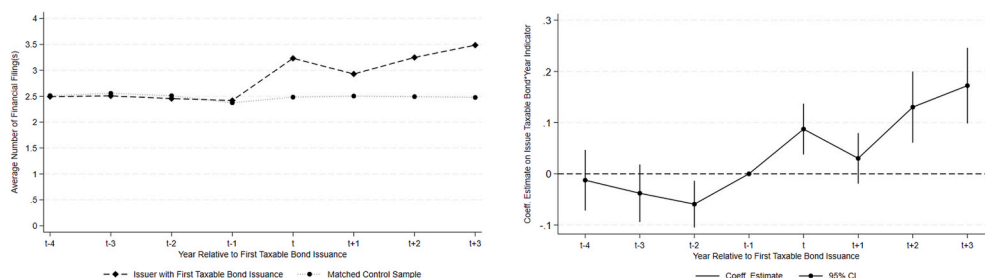


Fig. 3. Financial Filings Around the First Taxable Bond Issuance

Panel A: The Average Number of Financial Filings for Issuers with a First Taxable Bond Issuance and a Matched Control Sample

Panel B: Dynamic Treatment Effect Coefficient Estimates

Notes: Fig. 3, Panel A plots the average number of financial filings per fiscal year, centered around year t . For issuers with a first taxable bond issuance ($Issue\ Taxable\ Bond = 1$), year t is the fiscal year of the taxable bond issuance. The matched control sample includes up to five issuers that have never issued taxable bonds and are matched to the treated issuers based on bond purpose, issuer state, and the amount of outstanding debt, all measured in the year of the treated issuers' first taxable bond issuance. Panel B reports the coefficient estimates and 95 percent confidence intervals from an estimation of equation 2, where the dependent variable is $\#Financial\ Filings$, and where we interact each relative year with the treatment indicator, using year $t-1$ as the base year. The dynamic effects shown in Panel B relate to the results reported in Table 4.

Table 4

The effect of an issuer's first taxable bond issuance on the quantity of continuing disclosures, estimated using a stacked generalized difference-in-differences design.

DV =	Pred. Sign	(1)	(2)	(3)
		Coeff.	Coeff.	Coeff.
		(z-stat)	(z-stat)	(z-stat)
$Issue\ Taxable\ Bond * Post$	+	0.1178*** (5.71)	0.2584*** (8.37)	0.1611*** (8.81)
$Log(Bond\ Count)$		0.2001*** (10.52)	0.1081*** (4.28)	0.1716*** (10.15)
$Log(New\ Issue\ Amt)$		0.0044*** (3.45)	0.0486*** (25.86)	0.0185*** (16.44)
$New\ Callable\ Bond\%$		0.1222*** (5.13)	0.3487*** (11.77)	0.2029*** (10.07)
$New\ GO\ Bond\%$		-0.1036*** (-3.88)	-0.5368*** (-17.72)	-0.2624*** (-12.08)
$Log(Amt\ Due)$		0.0061*** (2.97)	0.0033 (1.05)	0.0060*** (3.20)
$Maturity$		-0.0032 (-0.44)	0.0020 (0.32)	-0.0011 (-0.18)
$Credit\ Rating$		-0.0034 (-0.32)	-0.0370*** (-2.71)	-0.0143 (-1.44)
$Credit\ Rating\ Missing$		-0.3190* (-1.69)	-0.8395*** (-3.59)	-0.4821*** (-2.74)
$\Delta Unemployment$		0.0049 (0.37)	-0.2089*** (-7.12)	-0.0436*** (-3.35)
ΔGDP		-0.0792 (-0.51)	-0.5470** (-2.02)	-0.1441 (-1.00)
<i>Fixed Effects</i>		Issuer-Cohort; Relative Year-Cohort	Issuer-Cohort; Relative Year-Cohort	Issuer-Cohort; Relative Year-Cohort
<i># of Observations</i>		111,188	111,188	111,188
<i># of Obs. Used in Poisson Estimation</i>		102,377	91,868	105,824
<i>Pseudo R²</i>		0.350	0.355	0.394

This table presents the results of a Poisson regression estimating equation (2). In column 1, the dependent variable is $\#Financial\ Filings$, defined as the number of financial filings made by an issuer during the fiscal year. In columns 2 and 3, the dependent variables are the number of event filings ($\#Event\ Filings$) and the total number of all continuing disclosures ($\#Continuing\ Disclosures$) made by an issuer during the fiscal year, respectively. $Issue\ Taxable\ Bond * Post$ is an indicator variable equal to one in the year of and the three years after an issuer's first taxable bond issuance. We expect the coefficient on $Issue\ Taxable\ Bond * Post$ to be positive, as indicated in the brackets. We include issuer-cohort fixed effects and relative year-cohort fixed effects. Standard errors are clustered at the issuer level. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively, using a one-tailed t -test where we denote a prediction, and a two-tailed t -test otherwise. Variables are defined in Appendix A.

preliminary visual evidence suggesting that treated and control issuers have similar disclosure levels and trends prior to the first taxable bond issuance. To further assess the reasonableness of the parallel trends assumption, we examine the dynamic effects of a first taxable bond issuance on issuers' disclosures. We re-estimate equation (2), interacting *Issue Taxable Bond* with separate indicators for each year (relative to the event year) over the estimation window (*Issue Taxable Bond** $t-4$, *Issue Taxable Bond** $t-3$, etc.). We use year $t-1$, the year immediately before the event year, as the benchmark (Barrios, 2022). We plot the coefficients on the interaction terms and the corresponding 95 percent confidence intervals in Fig. 3, Panel B. We find statistically insignificant coefficients on the interaction terms for years $t-4$ and $t-3$. Although the interaction term for year $t-2$ is negative and statistically significant (p -value < 0.05 , based on two-tailed tests), it is inconsistent with a pre-trend of increasing disclosure for treated issuers. Collectively, these results are broadly consistent with the parallel trends assumption being reasonable in our setting (Roberts and Whited, 2013). Focusing on the post-treatment years, we find that the coefficients on the interaction terms are positive and significant in all years except year $t+1$. Thus, the positive effect of first taxable bond issuances on issuers' disclosures is persistent, and issuers continue to provide more financial filings two to three years after the issuance of the first taxable bond.

4.4. Examination of an issuer's call of its last taxable bond

Next, we examine changes in continuing disclosures following the call of an issuer's last taxable bond. If issuers increase disclosure because taxable bonds attract a clientele that demands disclosure, we expect a decrease in disclosure once the issuer no longer has a taxable bond outstanding and experiences the corresponding tax clientele shift. However, prior research also suggests that disclosures are "sticky" (Einhorn and Ziv, 2008), such that issuers may not reduce disclosure after their last taxable bond is called. In other words, the disclosure response to the initiation and call of taxable bonds may not be symmetric.

Fig. 4, Panel A plots the average number of financial filings for issuers that called their last outstanding taxable bond and matched control issuers for years $t-4$ through $t+3$. We define the event year (year t) as the year in which an issuer's last taxable bond is called. Across all pre-treatment years (i.e., year $t-4$ to year $t-1$), treated issuers consistently provide more financial filings, on average, than control issuers. This pattern is consistent with our main finding that issuers with taxable bonds outstanding provide more disclosures. While there is a slight increase in financial filings for the control sample over time, we observe a small decline in financial filings in years $t+1$ and $t+2$ for the treated issuers. These changes, however, are small in magnitude.

Next, we estimate equation (3) and report the results in Table 5. In column 1, where the dependent variable is *#Financial Filings*, we observe a negative and significant coefficient on the interaction term *Call Taxable Bond***Post*. The coefficient estimate suggests that, on average, issuers decrease the number of financial filings by 5.4 percent after calling their last taxable bond, relative to the matched control sample. Using the mean number of 2.7 financial filings for the control issuers in the pre-period as the baseline (untabulated), the estimated percentage decrease implies 0.1 fewer filings. In column 2, where the dependent variable is *#Event Filings*, we observe a positive coefficient on *Call Taxable Bond***Post*, opposite to the prediction. This result is likely driven by material bond calls mechanically resulting in additional event filings. Consistent with the decrease in financial filings and increase in event filings having offsetting effects, we do not observe a statistically significant coefficient on the interaction term when the dependent variable is *#Continuing Disclosures* (column 3). Collectively, these results suggest that issuers adjust their continuing disclosures, particularly financial filings, in response to changes in their tax clientele. While our estimated decrease in financial filings is statistically significant, the economic magnitude of the decrease after issuers call their last taxable bonds is smaller than the increase in disclosure after the first taxable bond issuance, consistent with disclosure "stickiness" and adjustment costs.

With respect to the parallel trends assumption, Fig. 4, Panel A provides visual evidence suggesting that there are no pre-trends in disclosure between the treated and control issuers. To provide further evidence, we re-estimate equation (3), interacting *Call Taxable Bond* with separate indicators for each year relative to the event year (*Call Taxable Bond** $t-4$, *Call Taxable Bond** $t-3$, etc.), and plot the coefficients on the interaction terms and the corresponding 95 percent confidence intervals in Fig. 4, Panel B. We continue to use year $t-1$ as the benchmark. The coefficients in the pre-treatment years are mostly statistically insignificant, except for year $t-4$, for which the coefficient on the interaction term is positive and marginally significant (p -value < 0.10 , based on two-tailed tests). The evidence is largely consistent with the parallel trends assumption being reasonable in our setting.

In the post-treatment period, we find negative coefficients on the interaction terms for years $t+1$ to $t+3$. This pattern is directionally consistent with treated issuers reducing disclosure after their last taxable bond is called. However, only the coefficients for years $t+1$ and $t+2$ reach statistical significance at conventional levels based on one-tailed tests. Because more than half of the bond calls occur toward the end of our sample period (i.e., 2020 and beyond), the power to detect an effect, particularly for year $t+3$ in the post-treatment window, is likely reduced due to larger standard errors in the estimates.

4.5. Changes in disclosures following advance refunding transactions around the TCJA

To isolate the tax clientele effect on disclosure, it is important to control for differences between issuer-years with taxable bonds and those without, as these differences can affect the decision to issue taxable bonds and disclosure practices. In our primary analysis, we include issuer fixed effects and a host of variables to control for potential differences. To further corroborate our inferences, we take advantage of a recent tax law change, which resulted in a plausibly exogenous increase in taxable bond issuances.

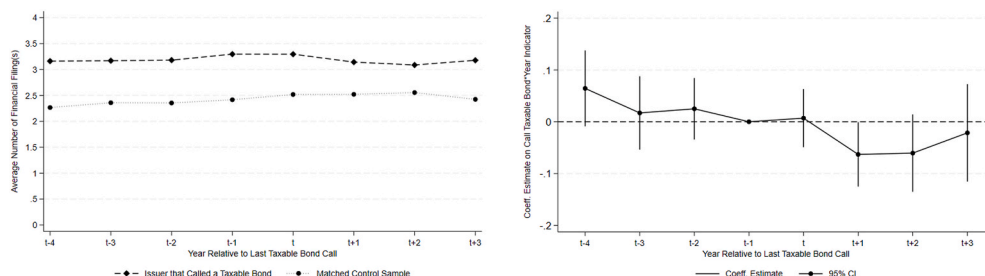
In response to interest rate changes or to restructure their debt portfolio, issuers can use refunding transactions to refinance an existing bond (Ang et al., 2017; Koutney et al., 2023; MSRB, 2016). In a typical refunding transaction, the issuer uses the proceeds from the issuance of a new bond (i.e., refunding bond) to fund a previously issued bond (i.e., refunded bond). When the refunded bond is not yet callable, the transaction is referred to as an advance refunding. Otherwise, the refunding transaction is referred to as a current

Table 5

The effect of an issuer's taxable bond call on the quantity of continuing disclosures, estimated using a stacked generalized difference-in-differences design.

DV =	Pred. Sign	(1)	(2)	(3)
		Coeff.	Coeff.	Coeff.
		(z-stat)	(z-stat)	(z-stat)
		<u>#Financial Filings</u>	<u>#Event Filings</u>	<u>#Continuing Disclosures</u>
<i>Call Taxable Bond*Post</i>	–	–0.0553** (–2.22)	0.1004 (2.27)	–0.0103 (–0.44)
<i>Log(Bond Count)</i>		0.1592*** (8.91)	0.1002*** (2.82)	0.1423*** (7.87)
<i>Log(New Issue Amt)</i>		0.0015 (0.87)	0.0564*** (21.32)	0.0179*** (11.71)
<i>New Callable Bond%</i>		0.1414*** (5.54)	0.3213*** (8.03)	0.1731*** (7.55)
<i>New GO Bond%</i>		–0.0553 (–1.62)	–0.6019*** (–11.95)	–0.2321*** (–7.53)
<i>Log(Amt Due)</i>		0.0079*** (3.04)	–0.0051 (–1.43)	0.0038* (1.65)
<i>Maturity</i>		0.0036 (0.43)	–0.0188 (–1.60)	–0.0040 (–0.51)
<i>Credit Rating</i>		–0.0017 (–0.16)	–0.0283* (–1.90)	–0.0107 (–1.19)
<i>Credit Rating Missing</i>		–0.2332 (–1.22)	–0.5963** (–2.43)	–0.3530** (–2.20)
<i>ΔUnemployment</i>		0.0146 (0.97)	–0.1152*** (–3.67)	–0.0189 (–1.27)
<i>ΔGDP</i>		–0.2311 (–1.19)	–0.6930** (–2.14)	–0.3488* (–1.93)
<i>Fixed Effects</i>		Issuer-Cohort; Relative Year-Cohort	Issuer-Cohort; Relative Year-Cohort	Issuer-Cohort; Relative Year-Cohort
<i># of Observations</i>		47,792	47,492	47,792
<i># of Obs. Used in Poisson Estimation</i>		43,018	39,259	44,809
<i>Pseudo R²</i>		0.324	0.349	0.373

This table presents the results of a Poisson regression estimating equation (3). In column 1, the dependent variable is *#Financial Filings*, defined as the number of financial filings made by an issuer during the fiscal year. In columns 2 and 3, the dependent variables are the number of event filings (*#Event Filings*) and the total number of all continuing disclosures (*#Continuing Disclosures*) made by an issuer in a fiscal year, respectively. *Call Taxable Bond * Post* is an indicator variable equal to one in the year of and the three years after an issuer calls its last outstanding taxable bond. We expect the coefficient on *Call Taxable Bond * Post* to be negative, as indicated in the brackets. We include issuer-cohort fixed effects and relative year-cohort fixed effects. Standard errors are clustered at the issuer level. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively, using a one-tailed *t*-test where we denote a prediction, and a two-tailed *t*-test otherwise. Variables are defined in [Appendix A](#).

**Fig. 4.** Financial Filings Around the Call of an Issuer's Last Taxable Bond

Panel A: The Average Number of Financial Filings for Issuers that Called their Last Outstanding Taxable Bond and a Matched Control Sample
Panel B: Dynamic Treatment Effect Coefficient Estimates

Notes: [Fig. 4](#), Panel A plots the average number of financial filings per fiscal year, centered around year *t*. For issuers with a taxable bond call (*Call Taxable Bond* = 1), year *t* is the fiscal year when an issuer's last taxable bond outstanding is called. The matched control sample includes up to five issuers that have never issued taxable bonds and are matched to the treated issuer based on bond purpose, issuer state, and the amount of outstanding debt, all measured in the fiscal year of the bond call. Panel B reports the coefficient estimates and 95 percent confidence intervals from an estimation of equation 3, where the dependent variable is *#Financial Filings*, and where we interact each relative year with the treatment indicator, using year *t*-1 as the base year. The dynamic effects shown in Panel B relate to the results reported in [Table 5](#).

refunding. The TCJA introduced a provision that prohibited the use of tax-exempt bonds for advance refunding transactions. Thus, in the post-TCJA period, issuers with an advance refunding transaction must issue a new *taxable* refunding bond, even if the refunded bond is tax-exempt. The TCJA setting offers an empirical benefit of examining disclosure changes associated with taxable bond issuances that are plausibly exogenous.

We examine changes in disclosure around advance refunding transactions completed in the post-TCJA period, relative to those completed in the pre-TCJA period. We estimate the following stacked difference-in-differences specification using a fixed-effects Poisson model:

$$\text{Disclosure} = \beta_1 + \beta_2 \text{Post TCJA} + \beta_3 \text{Post AR} + \beta_4 \text{Post TCJA} * \text{Post AR} + \Sigma \beta \text{Controls} + \text{Issuer Fixed Effects} + \text{Year Fixed Effects} + \varepsilon \quad (4)$$

For this analysis, we focus on advance refunding transactions and issuers who have executed such transactions both before and after the tax law change to hold constant the choice of refunding type.²⁹ The within-issuer comparison (i.e., comparing disclosure changes following the same type of transaction for the *same issuer* in the pre-versus the post-TCJA period), in particular, helps alleviate the concern that issuer-specific factors are correlated with both the choice of advance refunding and disclosure practices. If tax clienteles affect disclosure, we expect a larger increase in disclosure following an advance refunding transaction executed in the post-TCJA period (when the refunding bond is taxable), relative to the change in disclosure following an advance refunding executed in the pre-TCJA period (i.e., $\beta_4 > 0$).

We obtain data on advance and current refunding transactions from Bloomberg.³⁰ *Post TCJA* is an indicator variable set to one for advance refunding transactions completed on or after January 1, 2018, when the tax law requiring the use of taxable bonds for advance refunding became effective. *Post AR* is an indicator set to one for the year of an advance refunding transaction, which we designate as year *t*. For this analysis, we retain only the year before and the year of the refunding to minimize overlap between the post-refunding years for pre-TCJA transactions and the post-refunding years for post-TCJA transactions.³¹ The coefficient of interest is the interaction term *Post TCJA* * *Post AR* (i.e., β_4), which captures the differential change in issuers' disclosure following advance refunding transactions in the post-TCJA period (when issuers are required to use taxable bonds), relative to the change in disclosure following advance refunding transactions in the pre-TCJA period. We include control variables consistent with equation (2). We create cohorts based on the year of refunding and construct the regression sample by stacking each cohort in event time (Baker et al., 2022). Because equation (4) compares changes in disclosure following advance refunding in different time periods (i.e., across cohorts), we replace issuer-cohort fixed effects with issuer fixed effects. We also include year fixed effects (as opposed to the relative year-cohort fixed effects in equations (2) and (3)) to control for time trends in disclosure.

We first present evidence on the impact of the TCJA on taxable bond issuances. Using bond level data from Bloomberg, we plot the percentage of taxable refunding bonds issued for advance refunding transactions by calendar year in Fig. 5. Consistent with the TCJA requiring the use of taxable bonds for advance refunding transactions, the percentage of taxable bonds issued for advance refunding exhibits a sharp increase in 2018, when the tax law change became effective, and is nearly 100 percent in 2020 and 2021.³² For reference, we include the percentage of taxable bonds issued in current refunding transactions, which are not affected by the TCJA, and we find that the percentage remains low and stable throughout the sample period.³³

Table 6 presents the results of estimating equation (4). In column 1, when the dependent variable is *#Financial Filings*, the coefficient on *Post TCJA* * *Post AR* is positive but marginally significant.³⁴ In columns 2 and 3, where the dependent variables are *#Event*

²⁹ For example, Ang et al. (2017) find that financially constrained municipalities may face pressure to advance refund since it allows them to reduce short-term cash outflows. Given that the choice of advance versus current refunding is likely endogenous, we do not consider issuers with current refunding transactions an appropriate comparison group.

³⁰ Bloomberg data on advance refunding transactions are not well-populated for the early part of our sample. Thus, regression analyses using these data include fiscal years 2015 through 2022.

³¹ Consider an issuer with an advance refunding in 2016 (i.e., pre-TCJA) and another in 2018 (post-TCJA). With a test window from *t*-4 to *t*+3, the post-refunding period (i.e., *Post AR* = 1) for the 2016 transaction will include disclosures from 2016 to 2019. If issuers increase disclosure more after a post-TCJA advance refunding due to the issuance of a taxable bond, as predicted, the disclosure increase (in years 2018 and beyond) will also be captured in the post-refunding period for the 2016 transaction. This "contamination" will incorrectly attribute the disclosure increase due to the 2018 transaction to the 2016 transaction. In untabulated analyses, we find similar inferences (i.e., a positive and significant coefficient on *Post TCJA* * *Post AR*; *p*-value < 0.1 based on one-tailed tests) if we extend *Post AR* to include the year after the refunding, which allows more time to observe disclosure changes.

³² We note that there are tax-exempt bonds issued in advance refunding transactions in 2018 and 2019. This is likely the result of noise in the classification of advance refunding transactions, where refunding bonds are classified as multipurpose (e.g., a portion of the proceeds are used for advance refunding and the remaining proceeds are used for other purposes) and/or potential data errors.

³³ In an untabulated analysis, we also observe an increase in the average offering yield for advance refunding bonds as compared to current refunding bonds beginning in 2018, when issuers are required to issue taxable advance refunding bonds. In contrast, prior to 2018, the average offering yield is similar for both advance and current refunding transactions. These patterns in the average bond offering yield further corroborate that the TCJA introduced an exogenous increase in taxable bond issuance among advance refunding transactions.

³⁴ In untabulated analysis, we find a statistically significant increase (*p*-value = 0.023 using one-tailed tests) in *annual* financial statement filings following advance refunding transactions in the post-TCJA period relative to the pre-TCJA period. However, issuers do not increase interim financial filings and budget filings. These patterns likely explain why the coefficient in column 1 is only marginally significant.

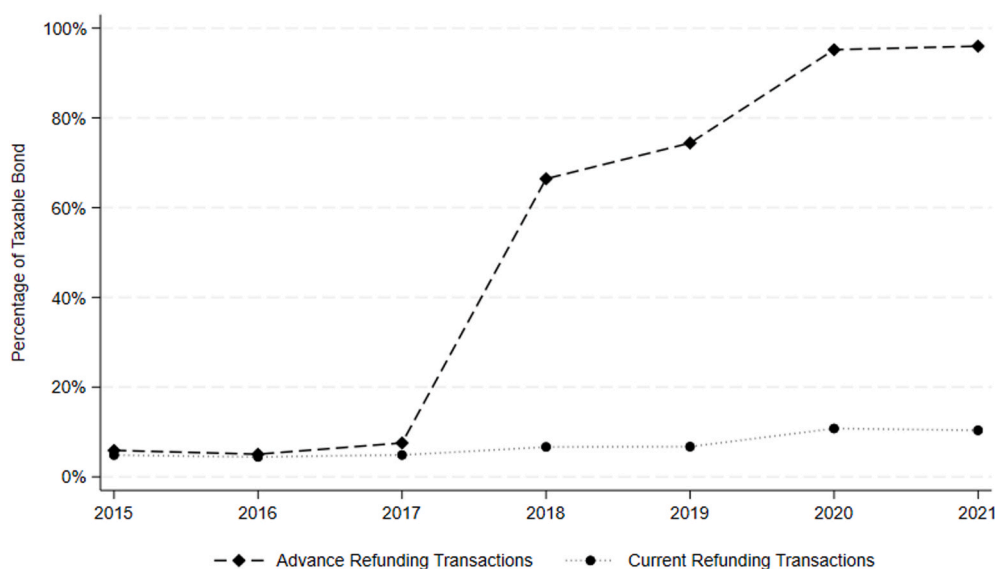


Fig. 5. The Percentage of Advance Refunding and Current Refunding Transactions with the Issuance of a Taxable Bond, by Calendar year
 Notes: This figure separately plots the percentage of taxable bonds issued in advance refunding and current refunding transactions, by calendar year, based on bond level data. We only include refunding transactions by issuers in our primary regression sample, and we restrict the sample period to 2015 and beyond.

Filings and *#Continuing Disclosures*, we observe a positive and statistically significant coefficient on *Post TCJA*Post AR*, consistent with an incremental increase in continuing disclosures following a plausibly exogenous increase in taxable bonds. Overall, these findings corroborate the primary analysis and are consistent with issuers' tax clienteles shaping their disclosure practices. However, we acknowledge that the sample size is relatively small when we restrict our analyses to issuers with an advance refunding in both the pre- and post-TCJA periods. We also caveat that we cannot fully address the potentially endogenous choice of engaging in advance refunding transactions.

5. Supplemental analyses and robustness tests

5.1. Timeliness of disclosure

Building on prior research showing that the rate of noncompliance with continuing disclosure requirements is high in the municipal bond market (e.g., Gillette et al., 2020; Maffett et al., 2025), our primary analyses focus on the amount of information provided by an issuer, measured by the number of disclosures. Given recent evidence that investors' reactions are stronger for timelier municipal disclosures (Cuny et al., 2025), in this section, we examine whether tax clienteles are associated with the timeliness of disclosure, another aspect of an issuer's disclosure practice. For this test, we focus on annual financial filings and measure timeliness using *Days to Financial Filing*, calculated as the number of days between the filing date for annual financial filings and the fiscal year-end associated with the filings. By construction, *Days to Financial Filing* decreases in disclosure timeliness. To alleviate the impact of potential outliers and noise in the data, we truncate *Days to Financial Filing* at zero and 366 (i.e., filings up to one year after the fiscal year-end).

We replace the dependent variables in equations (1)–(3) with *Days to Financial Filing* and present the results in Table 7. Columns 1, 2, and 3 report the results of estimating equations (1)–(3), respectively. If the clientele of taxable bonds demands timelier disclosure and issuers respond to such demand, we expect a negative coefficient on *Ind. Taxable Bond* and *Issue Taxable Bond*Post* in columns 1 and 2, respectively. If the response is symmetric, we expect a positive coefficient on *Call Taxable Bond * Post* in column 3. In column 1, we estimate a statistically insignificant coefficient on *Ind. Taxable Bond*. In column 2, we find a negative and significant coefficient on *Issue Taxable Bond*Post*, consistent with an increase in the timeliness of annual financial filings following an issuer's first taxable bond issuance. The coefficient estimate suggests that *Days to Financial Filing* decline by 1.3 percent, or by 2.5 days using 190.1 days—mean filing lag for the control group in the pre-period—as the baseline (untabulated). In column 3, the coefficient on *Call Taxable Bond*Post* is positive but not statistically significant at conventional levels. Collectively, Table 7 provides evidence of an increase in disclosure timeliness after the initial issuance of a taxable bond.

5.2. Other robustness tests and supplemental analyses, tabulated in Online Appendix

In cases of a count-based dependent variable, prior research commonly estimates an OLS regression using log-transformed dependent variables. However, Cohn et al. (2022) demonstrate that this approach can induce bias and create estimates that are difficult to interpret in relation to the untransformed count variable. Thus, we follow the recommendations of Cohn et al. (2022) and

Table 6

Estimation of differential changes in continuing disclosures following advance refunding transactions completed in the pre- and post-TCJA period.

DV =	Pred. Sign	(1)	(2)	(3)
		Coeff.	Coeff.	Coeff.
		(z-stat)	(z-stat)	(z-stat)
<i>Post TCJA</i>		<i>#Financial Filings</i> 0.0917 (0.68)	<i>#Event Filings</i> 0.0140 (0.12)	<i>#Continuing Disclosures</i> 0.0627 (0.53)
<i>Post AR</i>		−0.0709 (−1.41)	0.0252 (0.60)	−0.0361 (−0.91)
<i>Post TCJA * Post AR</i>	+	0.0827* (1.29)	0.1142** (1.90)	0.0908** (1.73)
<i>Controls</i>		Yes	Yes	Yes
<i>Fixed Effects</i>		Issuer; Year	Issuer; Year	Issuer; Year
<i># of Observations</i>		3710	3710	3710
<i># of Obs. Used in Poisson Estimation</i>		3710	3656	3710
<i>Pseudo R²</i>		0.830	0.564	0.814

This table presents the results of a Poisson regression estimating equation (4). *Post TCJA* is an indicator variable equal to one for advance refunding transactions completed on or after January 1, 2018. *Post AR* is an indicator variable equal to one for the year of an advance refunding transaction. In column 1, the dependent variable is *#Financial Filings*, defined as the number of financial filings made by an issuer during the fiscal year. In columns 2 and 3, the dependent variables are the number of event filings (*#Event Filings*) and the total number of all continuing disclosures (*#Continuing Disclosures*) made by an issuer in a fiscal year, respectively. The sample is restricted to the issuers with an advance refunding transaction in both the pre-TCJA and post-TCJA periods. Control variables are consistent with those used in estimating equation (2), untabulated for parsimony. We expect the coefficient on *Post TCJA * Post AR* to be positive, as indicated in the brackets. We include issuer fixed effects and year fixed effects. Standard errors are clustered at the issuer level. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively, using a one-tailed *t*-test where we denote a prediction, and a two-tailed *t*-test otherwise. Variables are defined in [Appendix A](#).

use a fixed-effects Poisson model. To evaluate the robustness of our results, we repeat analyses in [Tables 3–5](#) using log-transformed dependent variables and OLS estimation (tabulated in the Online Appendix, [Tables OA.1 to OA.3](#)). We observe consistent results using OLS estimation.

Our primary estimation of equation (1) includes an issuer fixed effect. Thus, issuers that always have or never have taxable bonds outstanding during our sample period do not directly contribute to our estimate of β_2 , but they indirectly affect β_2 through coefficient estimates on the control variables, which could induce bias ([Breuer and deHaan, 2024](#)). To alleviate such concern, we re-estimate equation (1) using a reduced sample of issuers that have variation in *Ind. Taxable Bond* (91,638 issuer-years and 7576 unique issuers). We repeat the analyses reported in [Table 3](#) using this subsample and find consistent results (tabulated in the Online Appendix, [Table OA.4](#)).

When constructing the primary sample, we make two notable sample selection decisions. First, for issuer-years without any

Table 7

The effect of taxable bonds on the timeliness of continuing disclosures.

DV =	(1)	(2)	(3)
	Coeff.	Coeff.	Coeff.
	(z-stat)	(z-stat)	(z-stat)
<i>Ind. Taxable Bond</i>	<i>Days to Financial Filing</i> 0.0004 (0.10)		
<i>Issue Taxable Bond * Post</i>		−0.0134** (−2.46)	
<i>Call Taxable Bond * Post</i>			0.0084 (1.05)
<i>Controls</i>	Yes	Yes	Yes
<i>Fixed Effects</i>	Issuer and Year	Issuer-Cohort; Relative Year-Cohort	Issuer-Cohort; Relative Year-Cohort
<i># of Observations</i>	323,762	83,090	35,432
<i>Pseudo R²</i>	0.533	0.534	0.574

This table presents the results of a Poisson regression estimating the difference in the timeliness of disclosures provided by municipal bond issuers with taxable bonds outstanding relative to those without (column 1), following the issuer's first taxable bond issuance (column 2), and following the call of an issuer's last outstanding taxable bond (column 3). The dependent variable, *Days to Financial Filing*, is the number of days between the filing date of issuer's earliest annual financial statement filed in fiscal year *t* and the fiscal year-end date associated with the filing. We truncate this variable at 0 and 366 to remove the impact of outliers and noise in the data. *Ind. Taxable Bond* is an indicator variable equal to one if the issuer had at least one taxable bond outstanding at the end of the fiscal year. *Issue Taxable Bond * Post* is an indicator variable equal to one in the year of and the three years after an issuer's first taxable bond issuance. *Call Taxable Bond * Post* is an indicator variable equal to one in the year of and the three years after an issuer calls their last outstanding taxable bond. We include control variables and the noted fixed effects. Standard errors are clustered at the issuer level. *, **, *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, based on two-tailed tests, respectively. Variables are defined in [Appendix A](#).

matched disclosure from MSRB, we reset all disclosure variables to zero. Second, if we cannot determine the fiscal year-end date for an issuer using matched financial disclosures from MSRB, we follow prior research and assign June 30, the most common fiscal year-end date for municipalities, as the fiscal year-end date (e.g., Gillette et al., 2020). These design choices are reasonable and allow us to retain the largest sample size. Nonetheless, we examine the robustness of our results using an alternative sample that excludes (1) issuers that do not have disclosures in MSRB during our sample period and (2) issuers for which we cannot determine the fiscal year-end date. Applying this alternative sample selection approach yields a sample of 418,168 issuer-year observations related to 41,535 unique issuers. Using this alternative sample, we repeat all analyses reported in Tables 3–5 (tabulated in the Online Appendix, Tables OA.5 to OA.7). We continue to find similar results.

Lastly, we examine whether the documented tax clientele effect on disclosure exists among subsamples of issuers for whom disclosure is mandatory and those for whom it is voluntary (i.e., issuers exempt from the requirement to file continuing disclosures under SEC Rule 15c2-12). If markets reward disclosure, issuers exempt from mandatory disclosure may be incentivized to *voluntarily* disclose. However, we do not have an ex-ante prediction as to whether the clientele effect on disclosure should differ between issuers subject to mandatory disclosure requirements and issuers exempt from such disclosure requirements. To the extent issuers subject to mandatory disclosure are already disciplined to comply with disclosure requirements, we may observe a smaller effect among these issuers. Yet, if regulatory efforts are ineffective (e.g., Maffett et al., 2025), we may observe significant effects among both issuers subject to the mandatory disclosure requirements and issuers exempt from disclosure requirements. We re-estimate equation (1) separately for issuer-years subject to continuing disclosure requirements and those *not* subject to such requirements (tabulated in the Online Appendix, Tables OA.8). We observe a positive relation between having a taxable bond outstanding and the disclosure level in both subsamples. Observing a tax clientele effect on *voluntary* disclosure is particularly interesting and further highlights the effectiveness of investor demand in shaping issuers' disclosure practices. An increase in voluntary disclosures among issuers not subject to continuing disclosure requirements is particularly interesting and further highlights the effectiveness of investor demand in shaping issuers' disclosure practices.

6. Conclusion

We examine whether the tax clienteles for municipal bonds affect issuers' disclosure practices. Compared to tax-exempt municipal bonds that attract high-tax retail investors, taxable bonds are more likely to attract a broader investor base, including institutional investors. If the taxability of municipal bonds shifts an issuer's tax clientele toward institutional investors, who are more likely to monitor and demand information, having taxable bonds outstanding should prompt municipal issuers to increase their continuing disclosures. Consistent with this tax clientele effect, our analyses suggest that municipal bond issuers provide more continuing disclosures in years with taxable bonds outstanding relative to years without. We also observe increases in continuing disclosures after an issuer's first taxable bond issuance and decreases in disclosures following an issuer's call of its last taxable bond. Exploiting a tax law change, we find increased disclosures after a plausibly exogenous increase in taxable bond issuances. Collectively, our results are consistent with the tax clienteles of municipal bonds shaping issuers' continuing disclosures. While existing research on municipal disclosure largely focuses on the lack of direct regulatory oversight as an explanation for disclosure noncompliance (e.g., Cuny et al., 2025; Gillette et al., 2020; Maffett et al., 2025), we show that the tax clientele effect, a market force, also shapes municipal disclosures. Our findings should be of interest to regulators, including the SEC and MSRB, who are concerned about the opaque disclosure environment in the municipal bond market.

Our study is not without limitations, most notably that we are unable to directly observe bond-level ownership and instead rely on the taxability of municipal bonds to capture an issuer's tax clientele. While various indirect descriptive statistics lend support to the validity of our approach, we acknowledge that we cannot provide direct evidence on bond ownership. Nevertheless, our approach of using bond taxability to infer clientele is meaningful and can be useful for future municipal bond research.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101910>.

Appendix A. Variable Definitions

Variable Name	Variable Definition
#Financial Filings	= The number of financial filings disclosed by issuer <i>i</i> in fiscal year <i>t</i> . Financial filings include, in brief, annual financial statements, quarterly financial statements, and separately filed budgets.
#Event Filings	= The number of material event notices filed by issuer <i>i</i> in fiscal year <i>t</i> . Material event notices under SEC Rule 15c2-12 include, in brief, notice of delinquencies, defaults, acceleration or termination events, unscheduled draws, adverse tax options, modifications to the rights of security holders or terms, material bond calls, defeasances, ratings changes, bankruptcy, insolvency or similar events, merger or acquisition activity, or a change in trustee.
#Continuing Disclosures	= The number of all filings made by issuer <i>i</i> in fiscal year <i>t</i> , including annual financial statements (audited or unaudited), other financial filings (e.g., budgets or quarterly filings), event filings (i.e., material event notices), and filings related to asset-backed securities.
Days to Financial Filings	= The number of days between the filing date of issuer <i>i</i> 's earliest annual financial statement filed in fiscal year <i>t</i> and the fiscal year-end date associated with the filing. We truncate this variable at 0 and 366 to remove the impact of outliers and noise in the data.
Ind. Taxable Bond	= Indicator variable equal to one if issuer <i>i</i> has at least one taxable bond outstanding at the end of fiscal year <i>t</i> .
Issue Taxable Bond	= Indicator variable equal to one if issuer <i>i</i> has a new taxable bond issuance during the year and does not have a taxable bond outstanding as of the end of the prior year (i.e., first taxable bond issuance).
Call Taxable Bond	= Indicator variable equal to one if issuer <i>i</i> called its last outstanding taxable bond during the year and does not have any taxable bonds outstanding as of the end of the year or thereafter.
Post	= Indicator variable equal to one for the year of and three years after the issuance of a first taxable bond or taxable bond call.
Post TCJA	= Indicator variable equal to one for transactions completed in 2018 or later.
Post AR	= Indicator variable equal to one in the year of an advance refunding transaction. Data compiled from Bloomberg.
Log(Bond Count)	= Natural log of one plus the number of outstanding bonds for issuer <i>i</i> in fiscal year <i>t</i> .
Log(New Issue Amt)	= Natural log of one plus the dollar value of new bonds issued by issuer <i>i</i> in fiscal year <i>t</i> .
Callable Bond%	= The dollar value of issuer <i>i</i> 's outstanding callable bonds divided by the dollar value of all issuer <i>i</i> 's outstanding bonds at the end of fiscal year <i>t</i> .
New Callable Bond%	= The dollar value of issuer <i>i</i> 's callable bonds issued divided by the total dollar value of issuer <i>i</i> 's newly issued bonds in fiscal year <i>t</i> .
GO Bond%	= The dollar value of issuer <i>i</i> 's outstanding general obligation (GO) bonds divided by the dollar value of all issuer <i>i</i> 's outstanding bonds at the end of fiscal year <i>t</i> .
New GO Bond%	= The dollar value of issuer <i>i</i> 's general obligation (GO) bonds issued divided by the total dollar value of issuer <i>i</i> 's newly issued bonds in fiscal year <i>t</i> .
Log(Amt Due)	= Natural log of one plus the principal amount (US\$) of bonds issued by issuer <i>i</i> that are due to mature in fiscal year <i>t</i> .
Maturity	= The shortest time to maturity (in years) of all outstanding bonds issued by issuer <i>i</i> at the end of fiscal year <i>t</i> .
Credit Rating	= The average long-term credit rating across all outstanding bonds for issuer <i>i</i> in fiscal year <i>t</i> . Missing ratings are reset to zero.
Credit Rating Missing	= An indicator variable set to one for each issuer <i>i</i> in fiscal year <i>t</i> where <i>Credit Rating</i> is missing.
ΔUnemployment	= The percentage change in state-level unemployment from fiscal year <i>t</i> -1 to <i>t</i> . Data was obtained from the Bureau of Labor Statistics. https://download.bls.gov/pub/time.series/la/
ΔGDP	= The percentage change in state-level GDP from fiscal year <i>t</i> -1 to <i>t</i> . Data was obtained from the Bureau of Economic Analysis. https://apps.bea.gov/regional/downloadzip.cfm

Filings are obtained from MSRB, and data on an issuer's outstanding bonds are obtained from Mergent.

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